Method of Transformations (Univariate) —> Another method to get the density function.

Note: We will only consider continuous random variables for this method.

Consider $U = h(Y)$ to be a function of the continuous random variable Y, with density $f_Y(y)$.

If $U = h(Y)$ is either increasing or decreasing (one-to-one) for all y (where $f_Y(y) > 0$) then $U = h(Y)$ has density

$$f_U(u) = f_Y\left(h^{-1}(u)\right)\left|\frac{d}{du}h^{-1}(u)\right|.$$

Note: If $u = h(y)$ then $y = h^{-1}(u)$ is the inverse function.

$u = h(Y)$, $f_Y(y)$

requirement: $h(Y)$ is increasing or decreasing.

$$f_U(u) = f_Y(h^{-1}(u)) \left| \frac{d\, h^{-1}(u)}{du} \right|$$

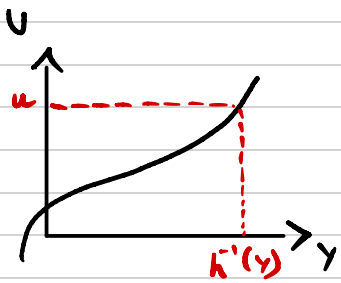
<u>Case 1</u> : Assume $h(Y)$ is an increasing function.

Follow the method of distribution.

$$P(U < u) = P(h(Y) < u) = P(Y < h^{-1}(u))$$

$$\parallel$$

$$G(h^{-1}(u))$$



$F(u) = P(U < u)$

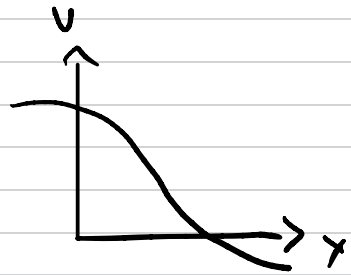
$G(y) = P(Y < y)$

$F(u) = G(h^{-1}(u))$

$$\frac{d}{du} F(u) = \frac{d}{du} G(h^{-1}(u))$$

$$= f_Y(h^{-1}(u)) \cdot \frac{d\, h^{-1}(u)}{du}$$

Case 2 : Assume $h(Y)$ is a decreasing function



Follow method of distribution

$$F(u) = P(U < u) = P(h(Y) < u)$$

$$P(Y > h^{-1}(u)) = 1 - P(Y < h^{-1}(u))$$

$$= 1 - G(h^{-1}(u))$$

$$\frac{d}{du} F(u) = \frac{d\,(1 - G(h^{-1}(u)))}{du} = -g_Y(h^{-1}(u)) \cdot \underbrace{\frac{d}{du} h^{-1}(u)}_{\text{But } \frac{dh^{-1}(u)}{du} < 0}$$

$$= -g_Y(h^{-1}(u)) \cdot -\frac{dh^{-1}(u)}{du}$$

$$= g_Y(h^{-1}(u)) \cdot \frac{dh^{-1}(u)}{du}.$$

Example 1:

The joint density of X and Y is given as

$$f(x,y)=\begin{cases}\dfrac{1}{\theta^2} & , \quad 0<y<\theta \text{ and } 0<x<\theta, \\ 0 & , \quad \textit{else}\end{cases}$$

Note: X and Y are independent $Uniform(0,\ \theta)$ random variables.

a) What is the probability density of $U=aY+b$?

b) What is the probability distribution of $U = -a\ln(Y),\ a > 0$?

Example 3:

Suppose $Z \sim Normal(0,1)$. $\Rightarrow f_Z(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}, \ -\infty < z < \infty.$

a) What is the probability distribution of $U = aZ + b,$ where $a > 0$?

b) What is the probability distribution of $U = Z^2$?